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Convertible wind energy based on predicted wind speed at hub-height

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ABSTRACT

Support vector machine is proposed to find wind speed at higher heights using measurements at lower heights. The mean absolute percentage error between measured and the estimated wind speed at height 40 m is found to be satisfactory. After validation at 40 m, the model was used to calculate the wind speed at hub heights up to 100 m. Annual energy yield was found to be increasing with hub height and, hence, accurate estimation of wind speed at heights becomes essential for realistic wind energy assessment. Furthermore, the plant capacity factor was found to be increasing approximately 1% for each 10-m increase in hub height.

KEYWORDS

Hub height; support vector machine; wind profile; wind speed

1. Introduction

The increasing pollution levels and adverse effect on climatic conditions have led policy makers to diversify sources of energy generation instead of sole dependence on fossil fuels. The renewable and clean sources of energy, such as wind, solar photovoltaic and thermal, geothermal, small and large hydro, biogas, tidal, etc., have achieved commercial acceptability and are being deployed in different parts of the globe. Of these, wind has become even economically at par with the regular fossil-fuel-based sources of energy generation. Due to commercial, economical, and environmental acceptability, wind power technology has been accepted as an alternative source of energy. At the end of June 2010, the cumulative global wind power installed capacity reached 175 GW (WWEA, 2011). Power of the wind is being used globally to meet small loads for remote applications (Notton et al., 2001) and is also being applied for hybrid power generation (Yang and Aydin, 2001).

Wind speed increases with height and, hence, it is crucial to estimate its values accurately with height at a potential site for wind farm development. However, wind speed measurements are not available at heights. Therefore, there is a need to extrapolate the WS up to the hub height based on measurements made at lower heights. Wind speed is the most important, critical, and fluctuating variable of the meteorological phenomena. It varies with time, location, and height and thus affects the wind shear profile (Spera, 1994). Furthermore, wind shear exponent (WSE) is significantly influenced by atmospheric stratification and changes with time (Genc et al., 2005; Dursun and Albayaci, 2011). Wind speed is also affected by ambient temperature and pressure and affects the performance of wind turbines (Elkhoury et al., 2010). Assareh et al., (2012) used artificial neural networks (ANNs) for the prediction of wind speed for Manjil, a city in Iran. İlkiliç and Nursoy (2010) presented the potential of wind energy for Turkey. Emami and Behbahani-Nia (2012) provided statistical evaluation of wind speed and power density in the Firouzkouh region in Iran.

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In the present work, a support vector machine (SVM) is used to estimate wind speed (WS) at different heights based on measurements at lower heights at Rowdat Ben Habbas (RBH), Saudi Arabia. It was observed that SVM outperforms several ANN models for prediction applications (Mohandes, 2002; Mohandes et al., 2004). Additionally, wind power at hub height from three chosen wind turbines was estimated using the extrapolated wind speed.

2. Support vector machine for regression

The major task of regression is to estimate a function $g(x)$ from given noisy samples of the function $G = \{(x_i, y_i)\}_{i=1}^N$. A SVM for regression applies a nonlinear mapping of the data x into a higher dimensional feature space and then performs a linear regression in this feature space (Vapnik et al., 1996):

$$f(x) = \sum_{i=1}^D w_i \Phi_i(x) + b, \quad (1)$$

where $\{\Phi_i(x)\}_{i=1}^D$ are features, b and $\{w_i\}_{i=1}^D$ are coefficients that need to be estimated from the data. The weights $\{w_i\}_{i=1}^D$ can be calculated from the data by minimizing the following function:

$$R[w] = \frac{1}{N} \sum_{i=1}^N |f(x_i) - y_i|_\varepsilon + \lambda \|w\|^2, \quad (2)$$

where λ is a regularization constant and the cost function defined by:

$$|f(x_i) - y_i|_\varepsilon = \begin{cases} |f(x_i) - y_i|_\varepsilon - \varepsilon & \text{for } |f(x_i) - y_i| \geq \varepsilon \\ 0 & \text{otherwise} \end{cases}, \quad (3)$$

is called Vapnik's ε -insensitive loss function. It is possible to show that the minimizing function has the following form:

$$f(x, \alpha, \alpha^*) = \sum_{i=1}^N (\alpha_i - \alpha_i^*) k(x_i, x) + b, \quad (4)$$

with $\alpha_i \alpha_i^* = 0$, $\alpha_i, \alpha_i^* \geq 0$ $i = 1, \dots, N$ and the kernel function $k(x_i, x)$ describes the inner product in the D -dimensional feature space:

$$k(x, y) = \sum_{j=1}^D \Phi_j(x) \Phi_j(y). \quad (5)$$

There is no need to calculate the features Φ_j ; instead, a kernel function is needed that has a known analytical form, like polynomial, radial basis function (RBF), and sigmoidal. The coefficients α_i, α_i^* can be calculated by maximizing the following function:

$$R(\alpha_i, \alpha_i^*) = -\varepsilon \sum_{i=1}^N (\alpha_i^* + \alpha_i) + \sum_{i=1}^N y_i (\alpha_i^* - \alpha_i) - \frac{1}{2} \sum_{i=1}^N (\alpha_i^* - \alpha_i) (\alpha_i^* - \alpha_i) k(x_i, x_i), \quad (6)$$

Subject to $\sum_{i=1}^N (\alpha_i^* - \alpha_i) = 0$, $0 \leq \alpha_i, \alpha_i^* \leq C$.

Only some of the coefficients α_i, α_i^* will have nonzero values, and the data points associated to them are defined as support vectors. The coefficients C and ε are user defined and need to be selected experimentally. The parameter b can be computed from all points that are on the margin. Further explanation of regression by SVM can be obtained from Vapnik et al. (1996).

3. Wind speed prediction at hub-height

Normally, WS measurements are made at 5–12 m above ground level (AGL). It is known that WS increases with height based on the following equation:

$$\frac{ws(h1)}{ws(h2)} = \left(\frac{h1}{h2} \right)^\alpha, \quad (7)$$

where $ws(h1)$ is WS at height $h1$, $ws(h2)$ is WS at height $h2$, α is the WSE and is reported to be equal to 1/7 in general. However, this value of α is not the same everywhere and at all times (Firtin et al., 2011). Using measured values of wind speed at different heights an accurate value of α (0.286) was obtained using a best fit exponential for RBH, Saudi Arabia (Rehman and Al-Abbadi, 2008). The present article uses WS data measured at RBH, a village inland of the Kingdom of Saudi Arabia located at latitude 29° 8.282' N, longitude 44° 19.817' E, and altitude 443 m. Measurements of wind speed at four heights (10–40 m) are recorded for 48 months between January 1, 2006 and December 31, 2009 by a high quality wind speed sensor NRG #40C maximum anemometer with three cups.

The SVM model is trained on measured WS values at heights between 10 and 30 m to extrapolate its value at 40 m. The trained SVM network is provided with WS values at heights 20 to 40 m to estimate its value at 50 m. Once the WS value at 50 m becomes available, it is used with the other measured values to estimate WS at 60 m. This process of using the measured and newly estimated values to estimate the WS at higher heights continues up to 100 m. To better assess the proposed method, average daily and hourly WS values are estimated.

3.1. Daily average wind profile

In the present work, initially a total of 1,460 records were considered but all records having WS values less than a threshold of 0.4 m/s were removed. Moreover, cases where WS at 10, 20, and 30 m was greater than WS at 20, 30, and 40 m, respectively, were removed. To validate the system, an SVM was trained using WS at 10 to 30 m to estimate values at 40 m. Thus, the measured data was divided into two parts, 80% for training and the remaining 20% for testing. For SVM implementation, a MATLAB environment was used and the RBF kernel was defined by $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$, where γ is a user defined positive parameter. All training and testing data were normalized between 0 and 1 to ease convergence. Several trials were made to select appropriate user specified parameters. The trial was started with ϵ equals 0.1, using grid search to select suitable values of C and γ . The values of γ and C were changed so that each will take one of the values in the sequence 2^{-6} , 2^{-5} , 2^{-4} , ..., 2^4 , 2^5 , and 2^6 . The pair (γ, C) , which gave the best performance on testing data, was selected. This process continued for a few cycles until final values of the parameters ($\gamma = 0.0015$ and $C = 1,600$) were selected. Next, several values of ϵ in the range of 0.1–0.4 were tried, however, it was noticed that the performance of the SVM on testing data was best at $\epsilon = 0.1$. The mean absolute percent error (MAPE) between the extrapolated and measured WS data at 40 m was found to be 1.58%. The performance of the proposed method was compared with the WSE-based estimate and 1/7th PL where MAPE was 1.5 and 4.81%, respectively.

The estimated and measured values for a certain number of records, for clarity, are shown in Figure 1a. Additionally, Figure 1b shows the scatter plot of measured and estimated values along with the best fit line passing through the origin with a coefficient of determination R^2 value of 0.99. Next, the WS values were estimated at 50 m. A new SVM was trained on all data between 10 and 30 m as inputs and WS at 40 m as output. The trained SVM was used to extrapolate the WS values at 50 m using values at 20, 30, and 40 m as inputs. Again, another SVM was trained using WS at 10, 20, 30, and 40 m as inputs and WS at 50 m as an output. This model was used to extrapolate the WS at 60 m using values at 20, 30, 40, and 50 m as inputs. This process continued by training a new SVM for each height to include the previously extrapolated values at lower heights along with the measured values to estimate WS at 60, 70, 80, 90, and 100 m.

The WS values estimated using SVM, local WSE = 0.286, and the 1/7th power law (PL) (WSE = 1/7 = 0.143) are compared in Figure 1c. The SVM estimated values are found to be in the middle between the other two methods. Table 1 shows the mean absolute difference between the SVM and

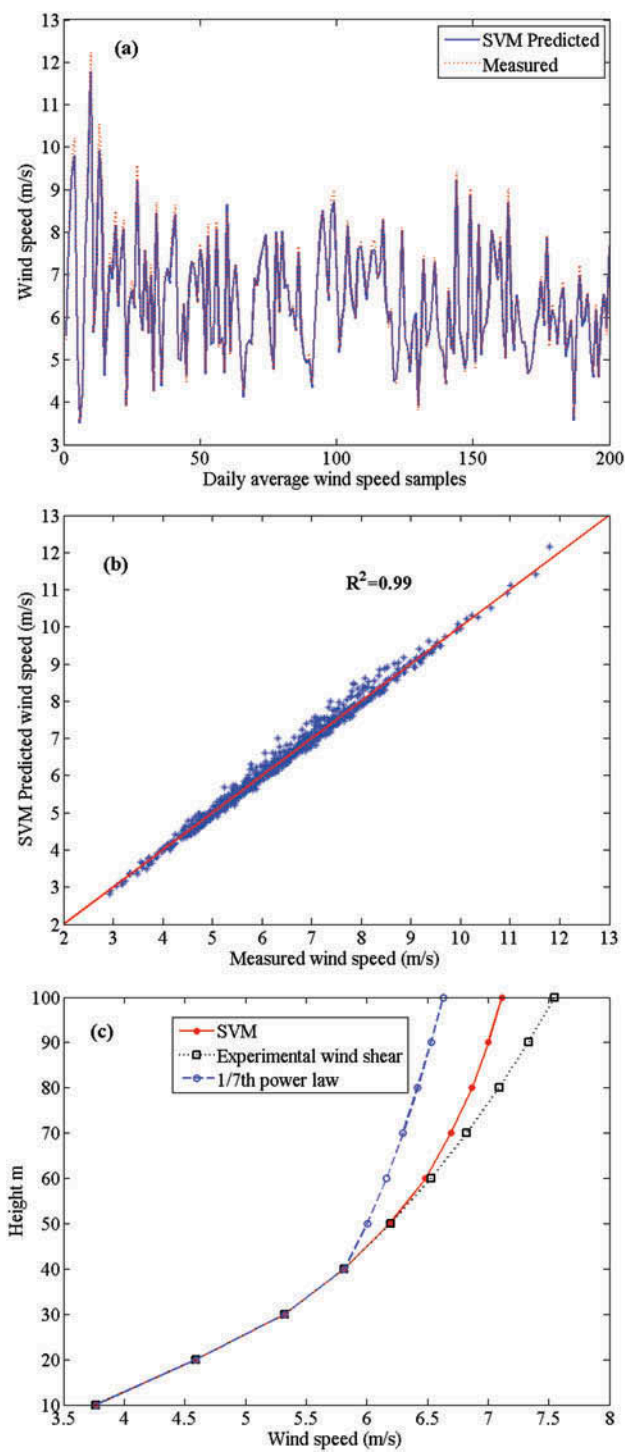


Figure 1. Results for daily average wind speed (a) at 40 m, (b) scatter plot at 40 m, and (c) performance of SVM between 50 and 100 m.

Table 1. Mean absolute percent differences between mean daily and mean hourly WS values predicted from the different methods.

Height, m	Daily Average			Hourly Average		
	1/7th PL and SVM, %	SVM and WSE, %	1/7th PL and WSE, %	1/7th PL and SVM, %	SVM and WSE, %	1/7th PL and WSE, %
50	2.75	0.45	3.21	4.15	0.87	3.24
60	4.77	1.14	5.96	6.30	0.31	5.97
70	5.42	2.28	8.35	7.12	1.13	8.33
80	7.87	2.30	10.35	7.34	2.87	10.42
90	8.17	2.65	11.03	7.22	4.74	12.30
100	10.29	3.24	13.86	6.96	6.43	13.84
Average	6.54	2.09	8.79	6.51	2.73	9.02

the 1/7th PL, SVM, and WSE, 1/7th PL and WSE at each height. The table indicates that the daily WS profile obtained from SVM is closer to the profile obtained from the WSE method.

3.2. Hourly average wind profile

Around 35,064 hourly values were filtered, as was done in the previous case, to obtain a total of 26,583 valid records. The valid records were divided into 80% for training and 20% for testing SVM with WS values at 10, 20, and 30 m as inputs and at 40 m as an output. Estimated WS at 40 m are shown to be in close agreement with measured values with a mean absolute percentage error (MAPE) of 1.91% and R^2 value of 0.987 as presented in Figures 2a and 2b, respectively. The extrapolated vertical profile at 50, 60, 70, 80, 90, and 100-m heights using 1/7th PL, SVM, and local WSE are shown in Figure 2c. Table 1 also summarizes the mean percent absolute difference between SVM and the 1/7th PL, SVM, and local WSE, and 1/7th PL and local WSE-based hourly estimations.

4. Convertible wind energy

Wind speed increases with heights and, hence, more energy can be obtained from wind turbines (Akpınar and Akpınar, 2006; Akdağ and Güler, 2011). This section deals with energy conversion using estimated wind speed at 60–100 m heights from 2-MW rated power wind turbines from three manufacturers (Weblink1, 2011; Weblink2, 2012; Weblink3, 2012). The wind power curves given in Figure 3 are used in conjunction with wind speed to estimate the monthly mean wind energy and plant capacity factor (PCF). The machines have a rotor diameter above 80 m and cut-in speed varied from 3.5–4 m/s corresponding to V90 from Vestas, Suzlon S.88, and AAER A84. The maximum annual energy yield of 4,889.83 MWh was found to be from Vestas V90 turbine at 100-m hub-height with PCF of 27.9%. The minimum annual yield of 4,551.28 MWh was obtained from AAER A84 with PCF of 26%, while Suzlon S.88 produced 4,686.89 MWh energy annually with a PCF of 26.8%.

The energy values presented in Figures 4 and 5 are the net values of energy delivered to the grid and are obtained by taking into consideration the overall losses of 11.51%, which included the down time losses of 3%, array losses of 5%, icing/soiling losses of 1%, and other losses of 3%. It is evident from Figure 4 that the maximum monthly mean energy was produced by Vestas V90 wind turbine, while the minimum was produced by AAER A84. In general, an increasing trend was observed in monthly mean energy values from January till June and then a decreasing trend was observed towards the end of the year with some exceptions. Almost the same seasonal trends were obtained for PCF values, as shown in Figure 5. Higher values of PCF of more than 30% were observed during February, March, April, June, and July and the minimum of about 20% was observed during September and November.

The energy yield and PCF increases almost linearly with hub height as can be observed from Figure 6. At a hub height of 60 m the Vestas-V90 wind turbine produced 4,266 MWh of electricity during the year, while at 100 m the annual energy yield was found to be 4,890 MWh. For each 10-m

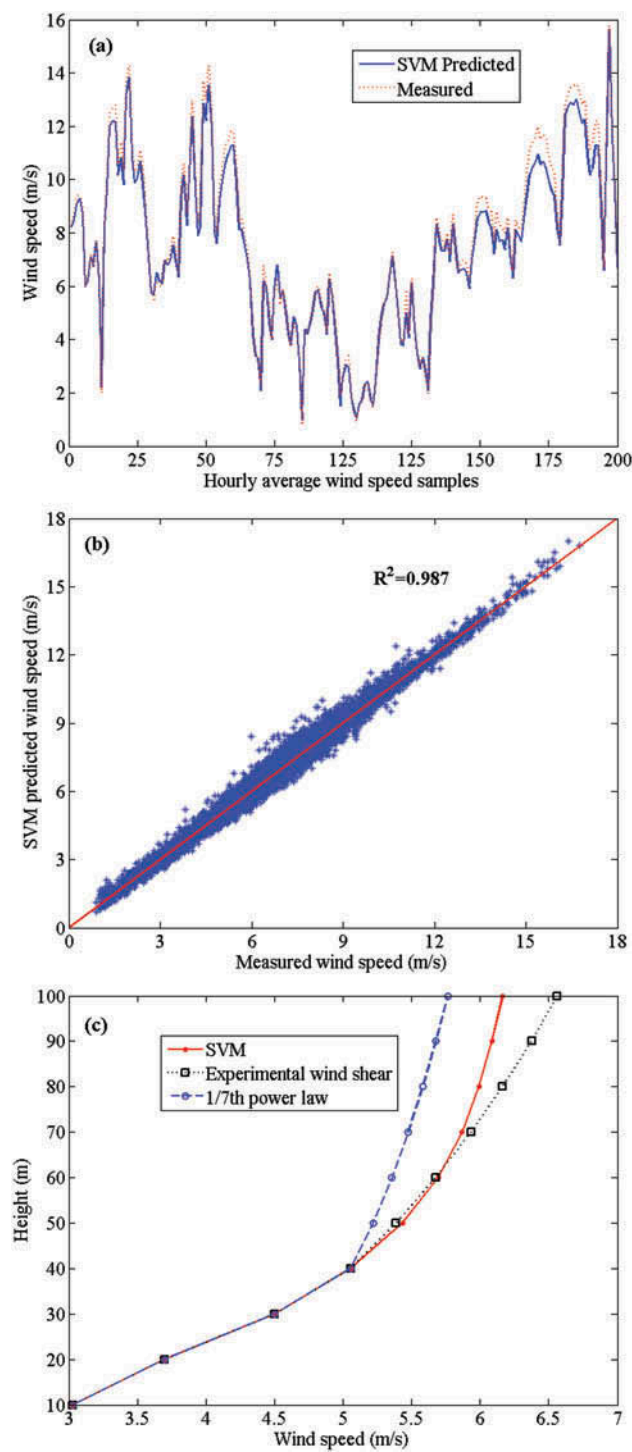


Figure 2. Results for hourly average wind speed (a) at 40 m, (b) scatter plot at 40 m, and (c) performance of SVM between 50 and 100 m.

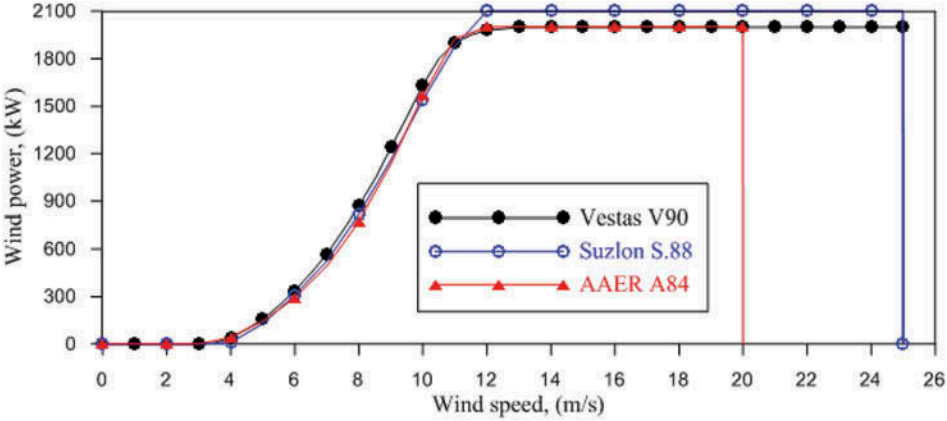


Figure 3. Wind power curves of wind turbines of 2-MW installed capacities.

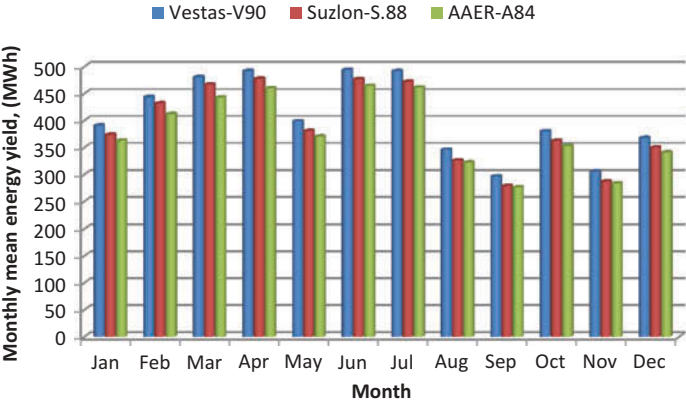


Figure 4. Monthly mean energy yield from 2-MW wind turbines.

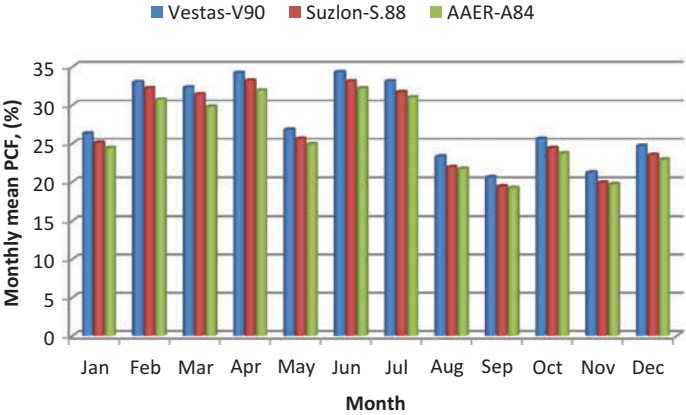


Figure 5. Monthly mean PCF of 2-MW wind turbines.

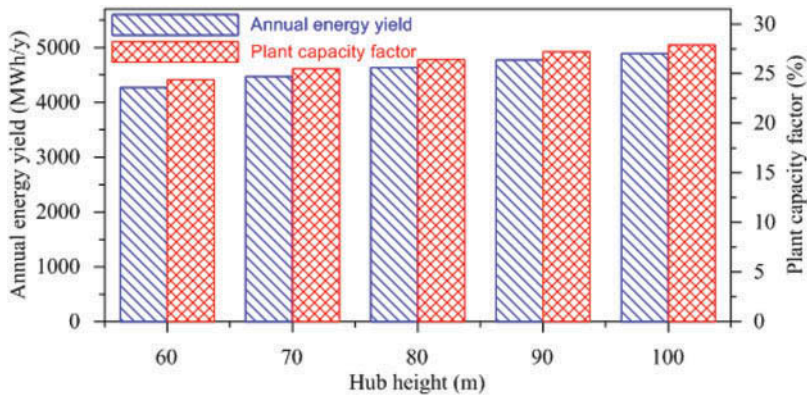


Figure 6. Monthly mean energy yield and plant capacity factor variation with hub height for a 2-MW wind turbine V90 from Vestas.

increase in hub height, almost 1% increase in PCF was obtained for Vestas-V90 wind turbine as can be seen from Figure 6.

5. Conclusion

This article showed that the SVM method is viable for extrapolating wind speed at higher heights from measurements recorded at lower heights. The method was validated by estimating known WS values at height 40 m using wind speed at heights between 10 and 30 m. The MAPE error between measured and estimated values was 1.58% for daily and 1.91% for hourly average. Moreover, the extrapolated (daily and hourly) average wind profiles were compared with the generated profiles using the 1/7th PL and the local WSE methods. Effect of hub height has been demonstrated on the energy yield and plant capacity factor. The annual energy yield increases almost linearly with hub height. The plant capacity factor was also found to be increasing by approximately 1% for each 10-m increase in hub height.

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Nomenclature

AGL	above ground level
ANN	artificial neural networks
SVM	support vector machine
WS	wind speed
WWEA	World Wind Energy Association
MAPE	mean absolute percent error
WSE	wind shear exponent
RBH	Rowdat Ben Habbas
RBF	radial basis function
PL	power law